

EXECUTIVE BRIEF: EMPLOYEE ATTRITION RISK ANALYSIS

A Data-Driven Retention Strategy for HR Leadership

1. Executive Summary

An analysis of IBM's HR database containing 1,470 employee records was conducted to understand the core drivers of employee attrition. Our findings indicate a baseline attrition rate of 16.12% across the organization. This rate is heavily imbalanced, meaning exits are concentrated within specific demographics, departments, and roles. By applying predictive analysis, we identified key operational and financial indicators that predict employee departures. Addressing these areas through targeted policy changes will prevent the costly loss of critical talent.

2. Core Drivers of Attrition (Top 3 Risk Factors)

- Overtime Work Status (Primary Predictor)

Employees working overtime are more than twice as likely to leave the company compared to their peers who do not work overtime. This suggests that work burnout and imbalance are primary drivers of exit.

- Salary and Financial Compensation

Attrition is heavily concentrated in lower salary brackets. Employees who left had a median monthly income of \$3,200, compared to \$5,204 for those who stayed. Nearly 33% of employees earning under \$2,500/month left the company, indicating standard salary thresholds are a clear factor in retention.

- Early Tenure and Onboarding Vulnerability

New hires represent a highly vulnerable group: 34.5% of employees in their first year at the company left. Attrition rates stabilize and drop below 15% only after the fifth year, emphasizing the need for robust onboarding support.

3. Critical Focus Areas: Departments & Job Roles

HR should immediately prioritize retention resources in the Sales department, which loses 20.6% of its staff. Specifically, the Sales Representative job role has a critical exit rate of 39.8%. The Research & Development department is more stable (13.8% attrition), but Laboratory Technicians (23.9%) and Research Scientists represent high-volume risk areas due to the sheer size of the department. Managers and Directors remain highly stable, with attrition below 5%.

4. Concrete Action Plan for HR Leadership

- Overtime Cap & Burnout Audits:** Implement a mandatory review policy for teams exceeding 10 hours of overtime per employee per week. Introduce flexible scheduling or compensatory time off, particularly for high-risk entry-level roles in Sales and Lab Tech.
- Targeted First-Year Engagement & Mentorship:** Establish a formal mentorship and buddy system for all new hires during their first 12 months. Conduct structured check-ins at 30, 90, and 180 days to address early tenure concerns.

5. Operational Limitations

While this predictive analysis provides a strong baseline, HR should note that it relies on static demographic and compensation data. It does not account for real-time sentiment, changes in manager relationships, team dynamics, or direct employee engagement feedback. Therefore, this tool should be used to identify high-risk groups for proactive conversations, rather than as the sole decision maker for promotions or individual career paths.